

Arduino-Based Dual-Mode SCIENTIFIC CALCULATOR

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This Arduino-based dual-mode scientific calculator is a versatile and efficient tool for both basic and advanced calculations. In normal mode, it functions as a standard calculator, performing addition, subtraction, multiplication, and division via a 4x4 keypad. Results appear clearly on an OLED screen. A push-button switch enables seamless toggling between modes for added flexibility.

In scientific mode, the calculator expands its functionality to handle advanced operations such as sine, cosine, tangent, and square root calculations—ideal for students, engineers, and professionals requiring trigonometric or algebraic computations. The OLED display ensures sharp visibility, while the keypad's multi-functional layout supports both standard and scientific inputs within a compact design.

The push-button mode switch improves usability, making the setup adaptable across varied use cases. The author's prototype is shown in Fig. 1, and the components required are listed in Bill of Materials table.

Circuit diagram and working

Fig. 2 shows the circuit diagram of the Arduino-based dual-mode scientific calculator. The design showcases efficient utilisation of hardware resources by combining an Arduino Uno, OLED display, and a simple keypad interface,

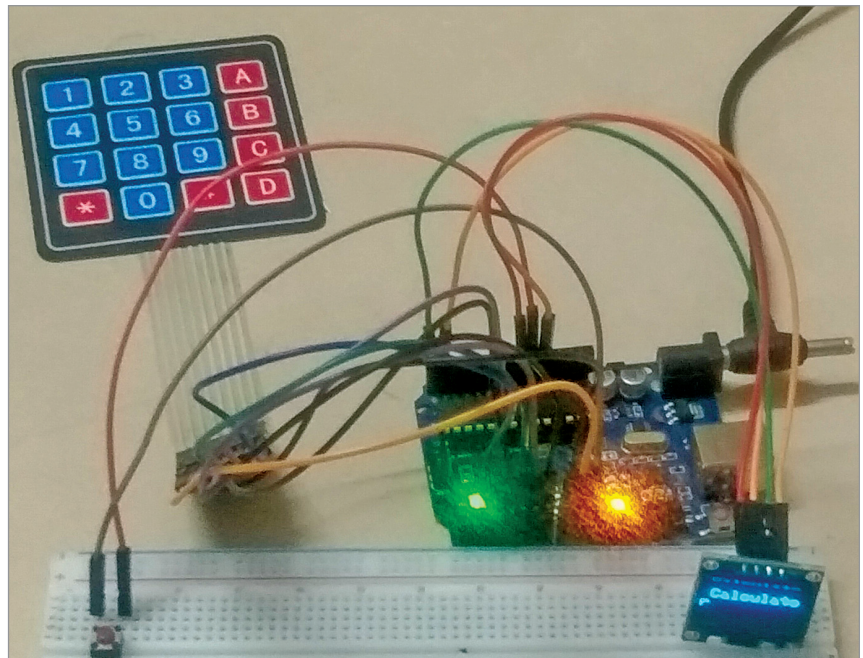


Fig. 1: Author's prototype

making the device cost-effective and straightforward to implement. The internal pull-up resistor used for the mode-switching button ensures stable operation by preventing false triggers.

This calculator also provides a valuable learning opportunity for electronics and programming enthusiasts. It covers essential concepts such as keypad interfacing, display handling, and implementing mathematical computations with Arduino. Overall, this customisable and expandable design offers a practical, hands-on introduction to embedded systems development.

The hardware setup includes

BILL OF MATERIALS

Item	Quantity
Arduino Uno	1
4x4 keypad	1
OLED display (SSD1306)	1
Push-to-on switch (SW1)	1
Jumper wires	As per requirement

connecting the OLED display via the I2C protocol (VCC to 5V, GND to GND, SDA to A4, and SCL to A5). The 4x4 keypad connects to the Arduino's digital pins, with rows attached to pins D9, D8, D7, and D6, and columns to pins D5, D4, D3, and D2. A push button is wired to ground and connected to pin D10